CityConnect

A real-time platform connecting citizens to timely civic action.

Team : CivicSync | Adani University |

Theme : Open Innovation

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IDEA

CityConnect

A unified platform for citizens and authorities to **manage complaints**, **track government projects**, and **navigate public transport** .

1. Civic Projects Visibility

- Ongoing and **upcoming** city projects by location
- **Clear** timelines, scope, and affected public areas
- Helps citizens **plan movement** and **avoid unsafe zones**

2. Complaint Reporting & Issue Management

- **Centralized civic complaint** submission
- **Community-driven** issue prioritization
- **Automated categorization** and severity tagging
- **Transparent tracking and status** updates

3. Public Safety & Alerts

- **Centralized** emergency alert system
- **Location-aware** safety notifications
- Authority-generated **real-time** updates
- **Preventive** awareness for vulnerable groups

4. Urban Mobility & Public Facilities

- **Unified** public transport planning
- **Cost- and time-optimized** routing
- **Multi-modal connectivity** insights
- Public property and facility discovery

System Architecture & Technology Stack

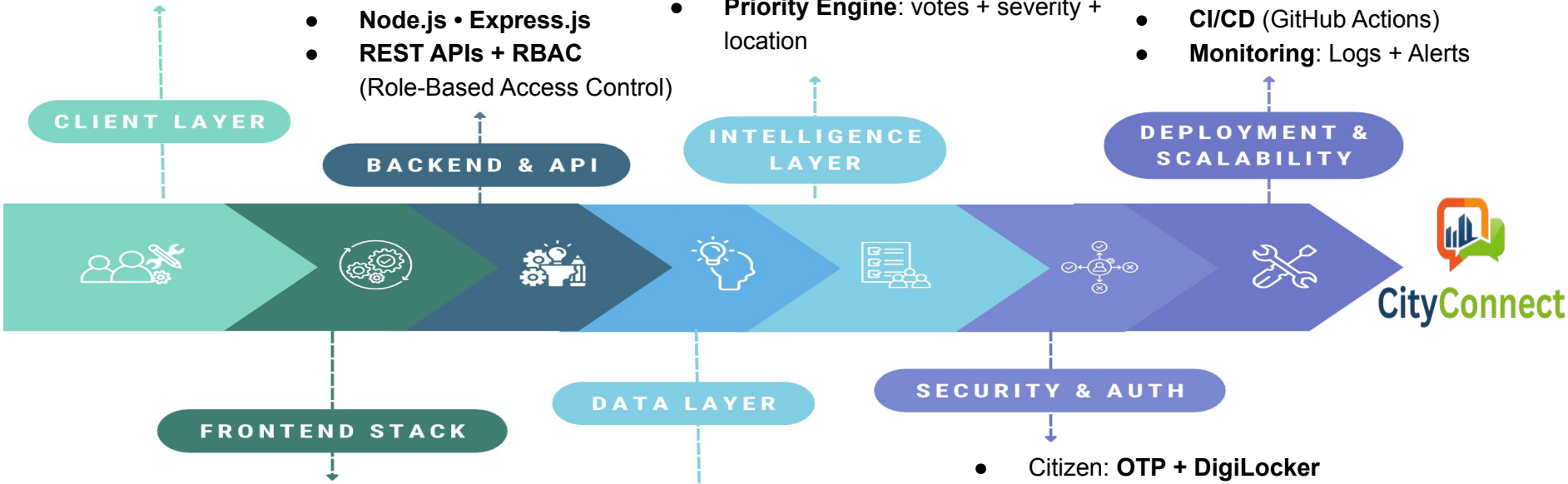
CityConnect

- Citizen • Govt
- Authority Dashboards
- Web + Mobile-responsive UI

- **Node.js • Express.js**
- **REST APIs + RBAC** (Role-Based Access Control)

- Complaint **Categorization & Severity Scoring**
- **Transport Routing**: Graph + Dijkstra/A*
- **Priority Engine**: votes + severity + location

- Docker • **Cloud** (AWS/Azure/NIC Cloud)
- **CI/CD** (GitHub Actions)
- **Monitoring**: Logs + Alerts



- **React (Vite) + React Router + Axios**
- UI: **Tailwind CSS**

- **PostgreSQL**
- ORM: **Prisma**
- Audit Logs + Status History

- Citizen: **OTP + DigiLocker**
- Govt: **SSO / Provisioned Accounts**
- JWT Sessions + RBAC

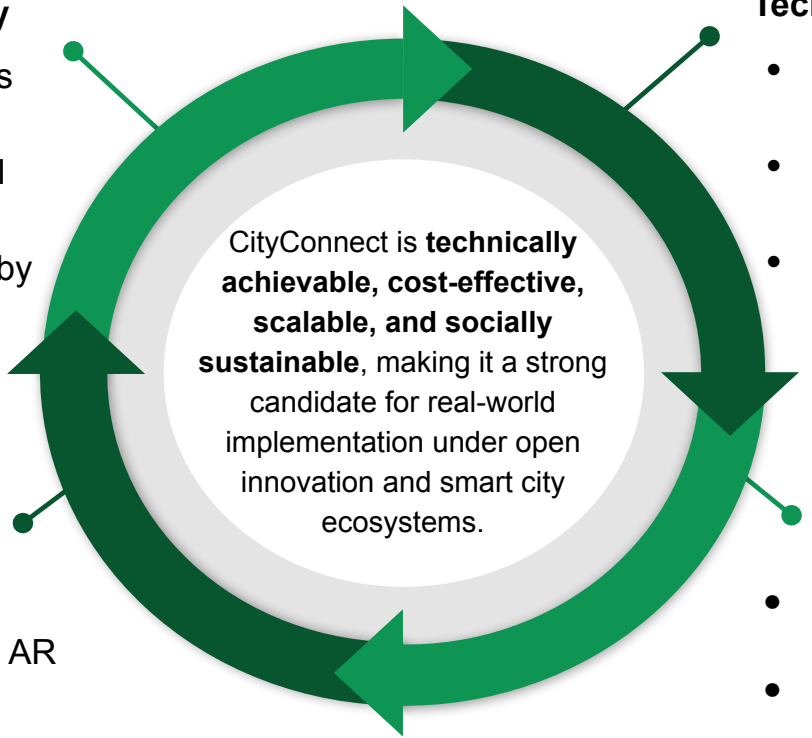
Feasibility & Viability

Operational Feasibility

- Easy adoption for authorities and citizens
- Automation reduces manual workload
- Multilingual and accessible by design

Technical Feasibility

- Uses existing AI/ML, cloud, APIs, and IoT technologies
- Modular design enables phased rollout
- Government system integration via secure APIs



Scalability & Sustainability

- Scalable across cities and regions
- Supports future AI, IoT, and AR upgrades
- Promotes long-term civic engagement

Economic Viability

- Low cost using open-source and cloud solutions
- Reduces long-term operational expenses
- Funding via Smart City programs and PPPs

CityConnect

Impacts & Benefits

Social Impact:

- Improves trust between citizens and government.
- Enhances safety with early alerts for riots, marches, or disasters.
- Encourages civic participation at scale.

Economic Impact:

- Optimizes city transport, reducing travel cost and time for millions.
- Transparent project progress reduces resource leakage and delays.

Environmental Impact:

- Promotes public transport use, reducing carbon emissions and congestion.

ALIGNMENT INCLUSIVITY AND ENPOWERMENTNMENTS

- Designed for all citizen groups, including women, elderly, low-income communities, and students.
- Multilingual support + voice inputs make it accessible to digitally weaker groups.

Citizens

- Gain a single platform to report issues, track solutions, check public facilities, and navigate the city easily.
- Empowered through upvotes, transparency, and real-time alerts for safety.

Government & Authorities:

- AI-based issue prioritization and clustering
- Real-time dashboards for faster response
- Data-driven planning with delay prediction & area insights

Daily Commuters & Students:

- Access the cheapest, fastest public transport routes; avoid disruption areas; and save time & cost.

CityConnect

References & Research

Key References

- **UN E-Government** Survey 2022
- **Smart City Mission** Framework (Gov. of India)
- **CPGRAMS** Architecture (DARPG India)
- IEEE Access – *Automated **Issue Categorization in Civic Complaints***
- **Dijkstra** (1959) – *Shortest Path Algorithm*
- **Elsevier** – *GIS-Based Route Planning for Urban Transport*
- **NDMA** – National Disaster Management Guidelines
- **Springer** – *Urban Data Platform Architecture*

Comparison Table (Our Solution vs Existing Methods)

Category	Existing Methods	Our Solution
Complaints Handling	Manual filing, scattered apps, slow updates	Unified platform, upvotes, real-time status
Transparency of Govt Projects	Info spread across PDFs/websites	Central dashboard with live timelines
Transport Guidance	Static routes, no cross-mode linking	Smart multimodal planner (Metro, BRTS, etc.)
Alerts & Notifications	SMS/news only, delays common	Instant city-level alerts from authorities
Data Integration	No combined view of civic data	Combined complaints, projects, transport, alerts
Scalability	Limited system-wide connectivity	Modular microservices + API-ready architecture

🟢 CityConnect – Live Working Prototype 🔗 https://cityconnect-nb5q.onrender.com